

Q1 2021 (01 Sep Shift 2)

If for the complex numbers z satisfying $|z - 2 - 2i| \leq 1$, the maximum value of $|3iz + 6|$ is attained at $a + ib$, then $a + b$ is equal to .

Q2 2021 (31 Aug Shift 2)

If z is a complex number such that $\frac{z-i}{z-1}$ is purely imaginary, then the minimum value of $|z - (3 + 3i)|$ is:

(1) $2\sqrt{2} - 1$

(2) $3\sqrt{2}$

(3) $6\sqrt{2}$

(4) $2\sqrt{2}$

Q3 2021 (31 Aug Shift 1)

A point z moves in the complex plane such that $\arg\left(\frac{z-2}{z+2}\right) = \frac{\pi}{4}$, then the minimum value of $|z - 9\sqrt{2} - 2i|^2$ is equal to .

Q4 2021 (27 Aug Shift 2)

Let z_1 and z_2 be two complex numbers such that $\arg(z_1 - z_2) = \frac{\pi}{4}$ and z_1, z_2 satisfy the equation

$|z - 3| = \operatorname{Re}(z)$. Then the imaginary part of $z_1 + z_2$ is equal to .

Q5 2021 (27 Aug Shift 1)

If $S = \left\{ z \in \mathbb{C} : \frac{z-i}{z+2i} \in \mathbb{R} \right\}$, then :

(1) S contains exactly two elements

(2) S contains only one element

(3) S is a circle in the complex plane

(4) S is a straight line in the complex plane

Q6 2021 (26 Aug Shift 2)

The least positive integer n such that $\frac{(2i)^n}{(1-i)^{n-2}}$, $i = \sqrt{-1}$ is a positive integer, is.

Q7 2021 (26 Aug Shift 2)

If $(\sqrt{3} + i)^{100} = 2^{99}(p + iq)$, then p and q are roots of the equation :

(1) $x^2 - (\sqrt{3} - 1)x - \sqrt{3} = 0$

(2) $x^2 + (\sqrt{3} + 1)x + \sqrt{3} = 0$

(3) $x^2 + (\sqrt{3} - 1)x - \sqrt{3} = 0$

(4) $x^2 - (\sqrt{3} + 1)x + \sqrt{3} = 0$

Q8 2021 (26 Aug Shift 1)

Let $z = \frac{1-i\sqrt{3}}{2}$, $i = \sqrt{-1}$. Then the value of

$21 + \left(z + \frac{1}{z}\right)^3 + \left(z^2 + \frac{1}{z^2}\right)^3 + \left(z^3 + \frac{1}{z^3}\right)^3 + \dots + \left(z^{21} + \frac{1}{z^{21}}\right)^3$ is.

Q9 2021 (26 Aug Shift 1)

The equation $\arg\left(\frac{z-1}{z+1}\right) = \frac{\pi}{4}$ represents a circle with:

(1) centre at $(0, -1)$ and radius $\sqrt{2}$

(2) centre at $(0, 1)$ and radius $\sqrt{2}$

(3) centre at $(0, 0)$ and radius $\sqrt{2}$

(4) centre at $(0, 1)$ and radius 2

Q1 (5)

$$|z - 2 - 2i| \leq 1$$

$$|x + iy - 2 - 2i| \leq 1$$

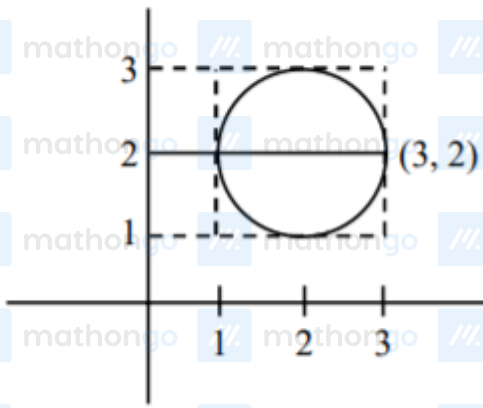
$$|(x - 2) + i(y - 2)| \leq 1$$

$$(x - 2)^2 + (y - 2)^2 \leq 1$$

$$|3iz + 6|_{\max} \text{ at } a + ib$$

$$|3i|z + \frac{6}{3i}|$$

$$3|z - 2i|_{\max}$$



From Figure maximum distance at $3 + 2i$

$$a + ib = 3 + 2i = a + b = 3 + 2 = 5 \text{ Ans.}$$

Q2 (4)

$\frac{z-i}{z-1}$ is purely Imaginary number

Let $z = x + iy$

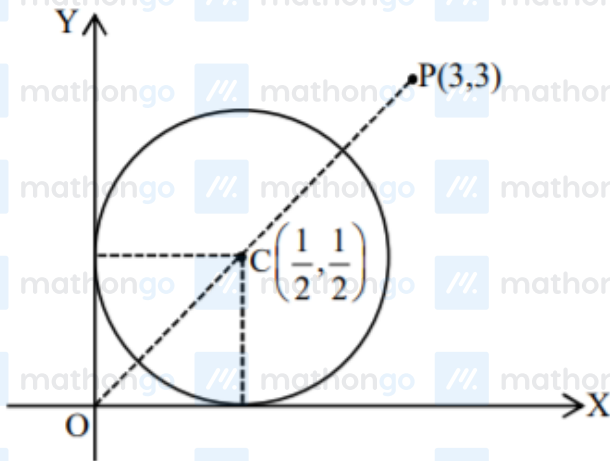
$$\therefore \frac{x+i(y-1)}{(x-1)+i(y)} \times \frac{(x-1)-iy}{(x-1)-iy}$$

$$\Rightarrow \frac{x(x-1)+y(y-1)+i(-y-x+1)}{(x-1)^2+y^2} \text{ is purely}$$

Imaginary number

$$\Rightarrow x(x-1) + y(y-1) = 0$$

$$\Rightarrow \left(x - \frac{1}{2}\right)^2 + \left(y - \frac{1}{2}\right)^2 = \frac{1}{2}$$



$$\begin{aligned}\therefore |z - (3 + 3i)|_{\min} &= |PC| - \frac{1}{\sqrt{2}} \\ &= \frac{5}{\sqrt{2}} - \frac{1}{\sqrt{2}} = 2\sqrt{2}\end{aligned}$$

Q3 (98)

Let $z = x + iy$

$$\arg\left(\frac{x-2+iy}{x+2+iy}\right) = \frac{\pi}{4}$$

$$\arg(x-2+iy) - \arg(x+2+iy) = \frac{\pi}{4}$$

$$\tan^{-1}\left(\frac{y}{x-2}\right) - \tan^{-1}\left(\frac{y}{x+2}\right) = \frac{\pi}{4}$$

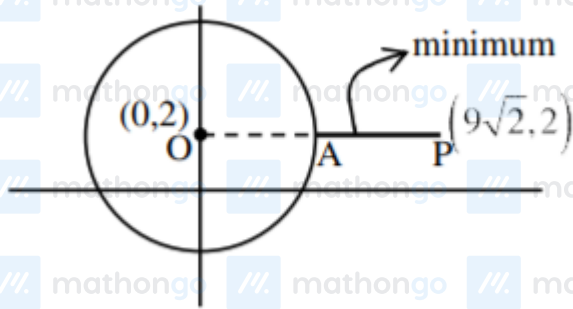
$$\frac{\frac{y}{x-2} - \frac{y}{x+2}}{1 + \left(\frac{y}{x-2}\right) \cdot \left(\frac{y}{x+2}\right)} = \tan \frac{\pi}{4} = 1$$

$$\frac{xy+2y-xy+2y}{x^2-4+y^2} = 1$$

$$4y = x^2 - 4 + y^2$$

$$x^2 + y^2 - 4y - 4 = 0$$

locus is a circle with center $(0, 2)$ & radius $= 2\sqrt{2}$



$$\begin{aligned}\text{min. value} &= (AP)^2 = (OP - OA)^2 \\ &= (9\sqrt{2} - 2\sqrt{2})^2 \\ &= (7\sqrt{2})^2 = 98\end{aligned}$$

Q4 (6)

$$|z - 3| = \operatorname{Re}(z)$$

$$\text{let } Z = x + iy$$

$$\Rightarrow (x - 3)^2 + y^2 = x^2$$

$$\Rightarrow x^2 + 9 - 6x + y^2 = x^2$$

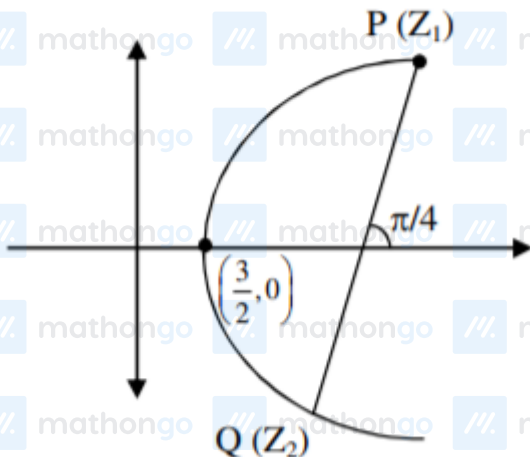
$$\Rightarrow y^2 = 6x - 9$$

$$\Rightarrow y^2 = 6\left(x - \frac{3}{2}\right)$$

$$\Rightarrow z_1 \text{ and } z_2 \text{ lie on the parabola mentioned in eq.(1)}$$

$$\arg(z_1 - z_2) = \frac{\pi}{4}$$

$$\Rightarrow \text{Slope of PQ} = 1.$$



Let $P\left(\frac{3}{2} + \frac{3}{2}t_1^2, 3t_1\right)$ and $Q\left(\frac{3}{2} + \frac{3}{2}t_2^2, 3t_2\right)$

$$\text{Slope of } PQ = \frac{3(t_2 - t_1)}{\frac{3}{2}(t_2^2 - t_1^2)} = 1$$

$$\Rightarrow \frac{2}{t_2 + t_1} = 1$$

$$\Rightarrow t_2 + t_1 = 2$$

$$\text{Im}(z_1 + z_2) = 3t_1 + 3t_2 = 3(t_1 + t_2) = 3(2)$$

Ans. 6.00

Aliter :

$$\text{Let } z_1 = x_1 + iy_1; z_2 = x_2 + iy_2$$

$$z_1 - z_2 = (x_1 - x_2) + i(y_1 - y_2)$$

$$\therefore \arg(z_1 - z_2) = \frac{\pi}{4} \Rightarrow \tan^{-1}\left(\frac{y_1 - y_2}{x_1 - x_2}\right) = \frac{\pi}{4}$$

$$y_1 - y_2 = x_1 - x_2 \quad (1)$$

$$|z_1 - 3| = \text{Re}(z_1) \Rightarrow (x_1 - 3)^2 + y_1^2 = x_1^2 \quad (2)$$

$$|z_2 - 3| = \text{Re}(z_2) \Rightarrow (x_2 - 3)^2 + y_2^2 = x_2^2 \quad (3)$$

sub (2) & (3)

$$(x_1 - 3)^2 - (x_2 - 3)^2 + y_1^2 - y_2^2 = x_1^2 - x_2^2$$

$$(x_1 - x_2)(x_1 + x_2 - 6) + (y_1 - y_2)(y_1 + y_2)$$

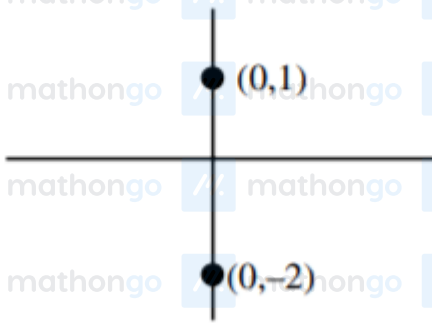
$$= (x_1 - x_2)(x_1 + x_2)$$

$$x_1 + x_2 - 6 + y_1 + y_2 = x_1 + x_2 \Rightarrow y_1 + y_2 = 6.$$

Q5 (4)

$$\text{Given } \frac{z-i}{z+2i} \in \mathbb{R}$$

Then $\arg\left(\frac{z-i}{z+2i}\right)$ is 0 or π



$\Rightarrow S$ is straight line in complex

Q6 (6)

$$\frac{(2i)^n}{(1-i)^{n-2}} = \frac{(2i)^n}{(-2i)^{\frac{n-2}{2}}}$$

$$= \frac{(2i)^{\frac{n+2}{2}}}{(-1)^{\frac{n-2}{2}}} = \frac{2^{\frac{n+2}{2}} i^{\frac{n+2}{2}}}{(-1)^{\frac{n-2}{2}}}$$

This is positive integer for $n = 6$

Q7 (1)

$$(2e^{i\pi/6})^{100} = 2^{99}(p + iq)$$

$$2^{100} \left(\cos \frac{50\pi}{3} + i \sin \frac{50\pi}{3} \right) = 2^{99}(p + iq)$$

$$p + iq = 2 \left(\cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3} \right)$$

$$p = -1, q = \sqrt{3}$$

$$x^2 - (\sqrt{3} - 1)x - \sqrt{3} = 0.$$

Q8 (13)

$$Z = \frac{1-\sqrt{3}i}{2} = e^{-i\frac{\pi}{3}}$$

$$z^r + \frac{1}{z^r} = 2 \cos \left(-\frac{\pi}{3} \right) r = 2 \cos \frac{r\pi}{3}$$

$$\Rightarrow 21 + \sum_{r=1}^{21} \left(z^r + \frac{1}{z^r} \right)^3 = 8 \left(\cos^3 \frac{r\pi}{3} \right) = 2 \left(\cos r\pi + 3 \cos \frac{r\pi}{3} \right)$$

$$\Rightarrow 21 + \left(z + \frac{1}{z} \right)^3 + \left(z^2 + \frac{1}{z^2} \right)^3 + \dots + \left(z^{21} + \frac{1}{z^{21}} \right)^3$$

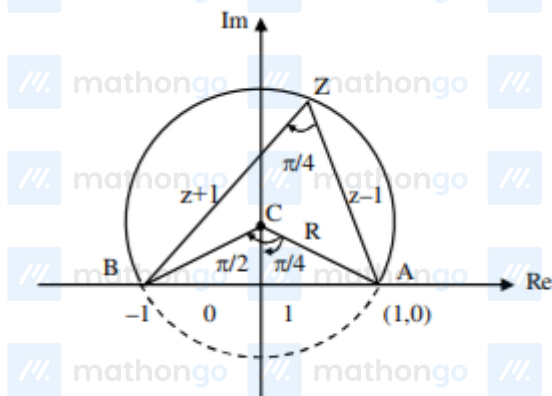
$$= 21 + \sum_{r=1}^{21} \left(z^r + \frac{1}{z^r} \right)^3$$

$$= 21 + \sum_{r=1}^{21} \left(2 \cos r\pi + 6 \cos \frac{r\pi}{3} \right)$$

$$= 21 - 2 - 6$$

$$= 13$$

Q9 (2)



In $\triangle OAC$

$$\sin\left(\frac{\pi}{4}\right) = \frac{1}{AC}$$

$$\Rightarrow AC = \sqrt{2}$$

$$\text{Also, } \tan \frac{\pi}{4} = \frac{OA}{OC} = \frac{1}{OC}$$

$$\Rightarrow OC = 1$$

$$\therefore \text{centre } (0, 1); \text{ Radius} = \sqrt{2}$$